Beyond Hallucination Detection: Measuring Consequential Uncertainty in LLM-Generated Code

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Abstract

Large language models generate code that looks uniformly confident. A correct call to `stripe.PaymentIntent.create()` and a deprecated call to `stripe.Charge.create()` are emitted with the same surface tone, even when the underlying token distribution is a near-50/50 split ($P = 0.52$ vs. 0.47). Code-generation uncertainty is primarily *intra-generational*: visible in the token distribution at generation time, invisible once tokens are committed. Sequence-level confidence signals — most notably the mean token probability p_{avg} of Spiess et al. [13] — read the same logprobs but aggregate them into one number per function; multi-sample methods measure output disagreement that simply does not occur for peaked code-token distributions, detecting at most 35% of HIGH-risk prompts on our benchmark at 5× cost.

We introduce ε, a per-token normalized Shannon entropy computed from the top-k log-probabilities returned by the generation API, AST-filtered to semantic tokens and max-pooled per function. At the function level ε is a recall-maximizing filter: on four models it flags 97.1% of HIGH API-integration prompts, with recall = 1.00 and precision = 0.272 on a 200-entry evaluated set. The contribution is not the function-level flag. The contribution is that the same score localizes: of the ~ 98 tokens in a typical flagged function, only 10.5% exceed the flag threshold and only 1.5% exceed the paused threshold. ε narrows the developer's review surface from a whole function to an average of 10.4 tokens at the flag threshold and 1.5 tokens at the paused threshold — a 89–98.5% reduction in within-function review burden. This is the distinguishing claim. p_{avg}, at the same 1× API cost, is competitive on raw detection (65–90% vs. ε's 70–85% across four models) and in fact beats ε on two of four OpenAI models; what it cannot do is say *where in the function* to look. ε answers both.

The low function-level precision is the design. We pair the filter with a parallel LLM review loop that reclassifies each flagged entry as CLEARED or CONFIRMED, using ε's token-level attribution to direct the reviewer to the specific 1–2 tokens that drove the score. On 201 entries across three models, a context-primed reviewer clears 68.7% of flagged entries with zero late false positives. The contribution is the two-level view: function-level recall plus token-level localization, together with the first direct three-way empirical comparison of intra-generational, sequence-level, and inter-generational uncertainty signals on production-realistic code prompts.

1 Introduction

AI-assisted developers now generate code several times faster than they can read it. A single engineer working with a coding assistant produces pull requests that would previously have represented days of work, and the bottleneck shifts from writing to reviewing. Reviewers have finite attention and cannot inspect every line with equal care, yet the failure modes of modern code generation

concentrate in small, hard-to-spot places. Recent empirical work on deprecated APIs quantifies the scale of the problem: LLM-based code completion emits deprecated API calls at rates of roughly 25–38% on standard prompts and substantially more on legacy codebases, because the training corpus straddles library-version transitions and the model has no posterior knowledge of deprecation at inference time [15].

A reviewer facing AI-generated code has two problems, not one. The first is *which functions* to scrutinize; the second is *where within each function* to look. Existing uncertainty signals can answer the first question: static analyzers catch known-bad patterns; sequence-level confidence scores such as the mean token probability p_{avg} [13] produce one number per function; multi-sample methods [10, 3, 12] re-roll the same prompt and measure output disagreement. None of them answer the second. On a function the size of a non-trivial SDK call — roughly a hundred tokens — the difference between "flagged" and "flagged here, on this token, between these two alternatives" is the difference between directing a reviewer and asking them to re-read everything.

A concrete instance. A model asked to charge a credit card might emit `stripe.Charge.create(...)` — deprecated since 2019 — or `stripe.PaymentIntent.create(...)`, the current API. Both arrive in the developer's editor with the same syntactic fluency, the same confident indentation, and no visible marker distinguishing a well-chosen call from one the model guessed at. At the method-selector token, the two choices split the probability mass near 50/50 ($P = 0.52$ vs. 0.47); the output exposes none of this. Failures of AI code generation cluster at exactly these points: version-split APIs, renamed SDK methods, signature changes between library releases. The model is uncertain; the output is not.

Intra- versus inter-generational uncertainty. The gap between a 0.52-vs-0.47 internal split and a confident-looking output is not an accident of any particular model; it is structural. Multi-sample uncertainty quantification methods for LLMs are *inter-generational*: they sample the same prompt multiple times and measure disagreement across completions [10, 3, 12]. This works when the model's uncertainty surfaces as instability across re-rolls — factual questions, open-ended reasoning. Code is not such a task. A model with a 0.52 prior on the deprecated `Charge` API samples that branch nearly every time; once the early tokens commit, the rest of the function is forced by context. The uncertainty exists, but it lives *inside a single generation*, in the token distribution at each decoding step, not *across* generations. We call this the *intra- versus inter-generational* distinction. It is the conceptual foundation of this paper.

ε, and two levels at which it operates. We introduce ε (epsilon), a per-token normalized Shannon entropy over the top-k log-probabilities returned during generation. After AST-based filtering to keep only semantically meaningful tokens (attribute accesses, keyword arguments, import targets — not whitespace, comments, or cosmetic identifiers), we aggregate via max-pooling to obtain a single score per function. The function-level score is a recall-maximizing filter: at threshold $\varepsilon \geq 0.30$ it flags 97% of HIGH prompts and catches 100% of true failures on our benchmark. A reviewer asked "which functions need scrutiny" gets a clear answer; at that scale, so does p_{avg}. But the function-level view discards most of what ε knows. Of the ~ 98 tokens in a typical flagged function, only 10.5% exceed the flag threshold and only 1.5% exceed the paused threshold. The same score that flags the function also tells the reviewer that the relevant uncertainty lives in the method selector on line 4, not in the parameter block on lines 7–12. ε is a two-level signal: a function-level filter that agrees with p_{avg} on "which," and a token-level localizer that no sequence-level aggregate can produce.

Figure 1 makes this concrete on Stripe Scenario A (§3.6): the function contains 98 semantic tokens, and the model's uncertainty concentrates at 2 of them. The reviewer does not need to re-read the function; it needs to look at two tokens.

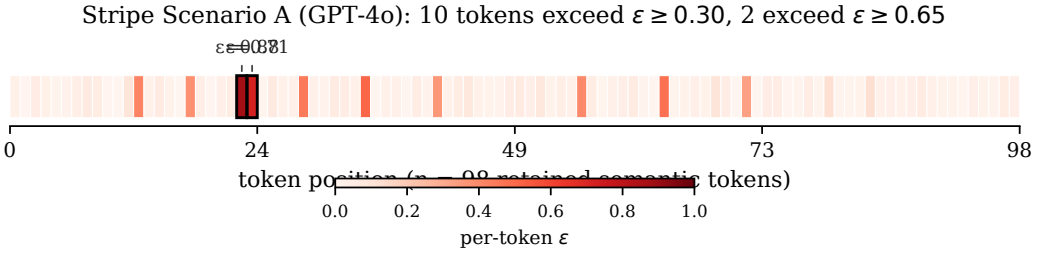

Figure 1: Of 98 semantic tokens in a typical flagged function (Stripe Scenario A, GPT-4o), ε exceeds the flag threshold at 10 tokens (10.2%) and the paused threshold at 2 tokens (2.0%). The function is flagged — but the within-function review surface is a small fraction of the generated code. Darker intensity = higher ε. See §3.6.

A three-way comparison. Because the function-level claim is not distinguishing, the comparison in this paper is between *signals*, not just detection rates. We benchmark ε against two alternatives at the same 1× API cost and one at 5× on the same production-realistic prompts across four models. At 1×: the sequence-level mean token probability p_{avg} [13] detects 65–90% of HIGH prompts — tying ε on GPT-4o, beating it on GPT-4o-mini and GPT-4-turbo, and losing to it on DeepSeek V3. p_{avg} deserves credit: on raw function-level detection it is competitive with ε and at the same cost. What it cannot do is identify the specific token at which the model was uncertain. ε produces both the function-level score and the token-level attribution. At 5× cost, multi-sample methods at $N = 5$, $T = 0.7$ detect 10–35% of HIGH prompts: the 52/47 hesitation does not surface as output variation, so inter-generational signals are structurally blind to it. UQLM [3], the closest packaged off-the-shelf tool, detects 5% on the same benchmark.

The two-stage system. A 27%-precision filter is not meant to be used in isolation. Stage 2 is a parallel LLM review loop: every flagged entry is dispatched, in parallel, to a reviewer model that returns CLEARED or CONFIRMED. The reviewer is not asked to re-read the whole function. It is asked to examine the specific high-ε tokens — on Stripe Scenario A, the 2 tokens whose ε exceeds the paused threshold. At current API prices this costs fractions of a cent per entry and completes in seconds. The token-level attribution is not decoration on the filter; it is the mechanism that makes the reviewer fast and precise. On 201 entries across three models, a context-primed reviewer clears 68.7% of flagged entries with zero late false positives.

Contributions.

1. The intra/inter-generational distinction, formalized and empirically verified: ε reaches 85% detection on GPT-4o and DeepSeek V3 against 10–35% for direct multi-sample measurement and 5% for UQLM on the same benchmark.
2. Token-level localization as ε's distinguishing claim: $\sim$ 98-token functions reduce to an average of 10.4 actionable tokens at the flag threshold (10.5% of the function) and 1.5 tokens at the paused threshold (1.5%) — an 89–98.5% reduction in within-function review burden. The function-level flag agrees with p_{avg}; the token-level attribution is what no sequence-level signal can produce.

3. p_{avg} credited: the sequence-level mean token probability of Spiess et al. [13] ties or beats ε on raw function-level detection across three of four models at the same $1\times$ API cost. ε's case rests on what p_{avg} cannot do, not on a rate difference.
4. A two-stage filter-plus-review architecture evaluated on 201 entries across three models (GPT-4o, GPT-4o-mini, DeepSeek V3): near-100% end-to-end precision at 100% recall, with the reviewer directed to ε's high-entropy tokens rather than asked to re-read the function.
5. The first direct three-way empirical comparison — ε vs. p_{avg} vs. multi-sample — on production-realistic version-split prompts, isolating where each signal fires and where it does not.

We use the labels HIGH and LOW throughout: HIGH prompts are API/version-split tasks across Stripe, OpenAI, SQLAlchemy, FastAPI, and Pydantic where an incorrect choice causes functional failure; LOW prompts are pure-logic tasks (sort, palindrome, Fibonacci, two-sum) where any correct algorithm is functionally equivalent. A detailed calibration table appears in §4.4.

Implementation and benchmark are at `https://github.com/michaeljaquith/epsilon-code`.

2 Related Work

Four adjacent lines of work inform the design of ε: empirical studies of deprecated API usage (which motivate the problem), token-level and sequence-level calibration of LLMs for code (which share the underlying logprob signal), multi-sample uncertainty quantification (which is the dominant alternative and the closest point of comparison), and adaptive decoding (which uses the same per-token entropy signal for a different purpose). We review each in turn, closing with static analysis.

2.1 Deprecated and version-split API usage

Wang et al. [15] present the first systematic study of deprecated API usage in LLM-based code completion, evaluating seven models on 145 API mappings drawn from eight popular Python libraries across 28,125 completion prompts. They measure API Usage Plausibility (AUP) and Deprecated Usage Rate (DUR), finding that deprecated-API emission is prevalent across all evaluated models; the root cause they identify is that deprecated API usages are present in training data while deprecation status itself is not reliably propagated to inference time. Their mitigations (REPLACEAPI and INSERTPROMPT) operate at the prompt or post-generation level and require a known deprecation mapping.

Their finding establishes that the problem we address is common in practice rather than a corner case: on version-split APIs and renamed SDK methods, models routinely commit to a deprecated branch without surface indication. Our work is complementary. Wang et al. measure *how often* LLMs emit deprecated code and diagnose *why* the output looks confident; ε measures the *signal at generation time* that distinguishes a confidently-chosen branch from an internally-contested one, and — crucially for the review loop — localizes that signal to a specific token rather than producing only a function-level score.

2.2 Token-level and sequence-level calibration

Spiess et al. [13] study calibration and correctness of language models for code, computing several confidence signals (including the sequence-level mean token probability, which we refer to throughout as p_{avg}) and evaluating whether they correlate with functional correctness on HumanEval and related benchmarks. They find that generative code models are not well-calibrated out of the box

and that Platt scaling can recover calibration in settings where labelled correctness data is available. The p_{avg} signal — the exponential of the mean log-probability over all generated tokens — is derived from the same raw logprob data that ε uses, and is the closest signal-level comparator for the function-level part of ε's output.

Four differences between that setup and ours deserve emphasis. First, *aggregation*: p_{avg} takes a sequence-level mean of per-token probabilities; ε takes a max after AST filtering, so a single high-entropy API-selector token is not diluted by long confident stretches. Second, *filtering*: ε restricts the input set to semantically meaningful tokens (attribute accesses, keyword arguments, import targets), while p_{avg} averages over every token including whitespace and cosmetic names. Third, *prompt regime*: Spiess et al. evaluate on HumanEval and similar benchmarks where correctness is well-defined by unit tests; we evaluate on version-split API prompts where the ground truth is ambiguous enough to require LLM judging (§6.1). Fourth, *output granularity*: p_{avg} produces one scalar per function and is silent on *which token* carried the uncertainty. This fourth distinction is the one that matters for the review loop (§5). Because p_{avg} and ε operate on the same underlying logprobs at the same $1\times$ API cost, a direct head-to-head comparison is possible; we report it in §5.

Earlier token-level calibration studies (Kadavath et al. [9]; Varshney et al. [14]) establish that LLMs are reasonably calibrated at the token level for factual questions and that selective generation can exploit this. Our work extends that line to code, where AST filtering is what makes per-token entropy actionable rather than dominated by cosmetic variance. Conformal prediction approaches [11, 1] provide set-valued predictions with coverage guarantees but do not localize which token within a generation is uncertain. Uncertainty propagation through multi-stage ML pipelines is studied in adjacent domains [8].

2.3 Multi-sample and behavioural-diversity methods

Inter-generational methods sample the same prompt multiple times and measure disagreement among outputs. Kuhn et al. [10] introduce semantic entropy: multiple completions are clustered by semantic equivalence (via natural language inference), and cross-cluster entropy provides a hallucination signal for factual QA. UQLM [3] packages semantic negentropy and related signals as an open-source library; we include it here as the closest off-the-shelf tool and report its performance in §5 in one sentence. Beyond these, a growing line of work [12] applies stronger notions of equivalence — up to and including symbolic execution or semantic-equivalence checks on generated code — to improve behavioural consistency detection specifically for code outputs.

These methods are structurally limited for the task we study. Intra-generational uncertainty does not surface as output variation when the distribution is peaked (as it typically is for code tokens: a 0.52-peaked distribution picks its peak nearly every time at temperature 0.7, and always at temperature 0). A model can be internally torn between two API calls and emit one of them unanimously across re-rolls. The Stripe example we report empirically in §5.3 — GPT-4o with $\varepsilon = 0.878$ and sample diversity $= 0.00$ across $N = 5$ at $T = 0.7$ — is a direct instance. A stronger equivalence check improves precision on Type 3 (implementation-approach) false positives, where two semantically equivalent implementations would be conflated by output-level diversity alone; but it cannot recover a signal that never varies across samples in the first place. The intra/inter distinction is not a claim that inter-generational methods are badly implemented, only that they are measuring a different quantity.

2.4 Generation-time intervention and adaptive decoding

AdaDec [6] reads the same per-token Shannon entropy that ε reads, but uses it at decoding time rather than review time: when the entropy at the current step exceeds a learned threshold, the decoder pauses and triggers a lookahead re-ranking step over candidate continuations. On HumanEval and MBPP across model sizes from 0.6B to 8B, AdaDec reports up to 15.5pp improvement in Pass@1 over greedy decoding while pausing on 3.95–11.88% of tokens — the same order of magnitude as ε's flagged-token rate of 10.5% (§3.6), which is not a coincidence: both identify the same set of decision points. The signal is the same as ours; the action is different. AdaDec improves the model's own output at generation time; ε surfaces residual uncertainty *after* the model has committed, for routing to downstream review. The two approaches are compatible: an AdaDec-enhanced generator still produces a token distribution at each step, and ε operates on whatever distribution the final decoder emits. We do not evaluate a combined pipeline but expect ε values to be lower on average after AdaDec-style intervention, with residual high-ε tokens concentrating on consequential decisions that even a lookahead re-rank could not resolve.

2.5 Static analysis, linters, and execution-based evaluation

Static analysis and linter rules catch *known*-bad patterns for specific libraries once deprecation has been encoded into a ruleset. These approaches are complementary to ε and represent the floor of detection: when a rule exists, a linter will catch the problem whether or not the model hesitated. ε operates in the gap — it fires where the model itself was uncertain, which includes version splits not yet in any ruleset (a library updated last week, a half-migrated codebase) and internal renames that static analysis has no fact to test against. Execution-based code-generation benchmarks [5, 2, 7] measure whether the output passes test cases, not whether the model was confident; a silent dependency update can turn a fully-passing generation into a regression that no test in the current suite exercises. To our knowledge, no prior work treats within-generation *token-level* uncertainty as a first-class routing signal for code review that localizes where within a function to look; the closest is CodeT [4] and related self-consistency approaches, which are inter-generational by construction, and post-generation repair with uncertainty-based ranking [16], which operates on already-generated candidates.

3 ε: Definition, Construction, and Localization

3.1 Per-token entropy

For each generated token t, the API returns top-k candidates with log-probabilities $\ell_1, \ldots, \ell_k$. Let $p_j = \exp(\ell_j)/\sum_j \exp(\ell_j)$ be the renormalized distribution. Define

$$\varepsilon_t \;=\; -\frac{1}{\log k}\sum_{j=1}^{k} p_j \log p_j \;\in\; [0,1]. \tag{1}$$

Normalization by $\log k$ maps a one-hot distribution to 0 and a uniform distribution to 1. We use $k = 5$ (the OpenAI cap), which covers the dominant mass for code tokens; tails beyond top-5 are negligible for the binary flag decision. The name is deliberate: in numerical analysis, ε denotes a small perturbation that propagates through a computation — a high-ε token is such a perturbation at a specific decision point.

Terminology: high and low prompts. We use the labels HIGH and LOW throughout. HIGH prompts are API/version-split tasks (Stripe, OpenAI, SQLAlchemy, FastAPI, Pydantic) where an incorrect choice causes functional failure. LOW prompts are pure-logic tasks (sort, palindrome, Fibonacci, two-sum) where any correct algorithm is functionally equivalent. The split and detection rates per category appear in §4.4.

3.2 Three types of code-generation uncertainty

Raw per-token entropy, applied naively, produces too many false positives to be useful: the bulk of high-entropy tokens in generated code are not risk indicators. Understanding which ones are, and why, is what drives the filtering design in the rest of this section. Empirical analysis of our calibration benchmark reveals three categorically distinct types of token-level uncertainty in generated code.

Type 1 — Cosmetic (naming). Uncertainty about what to *call* something (`sort_list` vs. `sort_numbers`). ε reaches 0.7–0.9, but any choice is correct. Dominates LOW prompts: the top flag on 9 of 10 LOW prompts was a function or parameter name on the first line.

Type 2 — Consequential (API/library decisions). Uncertainty about which *API or library* to use. Example: `stripe.Charge.create()` (prob. 0.52) vs. `stripe.PaymentIntent.create()` (prob. 0.47), $\varepsilon = 0.73$ at the attribute token. This is the target signal.

Type 3 — Implementation-approach. Uncertainty among multiple correct algorithms (recursive vs. iterative Fibonacci, for-loop vs. comprehension, brute vs. hash-based two-sum). *The Type 2 / Type 3 distinction is definitional, not structural.* A given high-ε token is classified as Type 3 precisely when all of the competing continuations produce functionally correct code; it is classified as Type 2 when at least one continuation is wrong. The underlying decoder state can look identical in both cases — a 0.5-peaked top-2 distribution at a decision point — and ε has no way, from the token distribution alone, to know which label applies. The label is a property of the external world (are all branches correct?) and is only visible via ground truth. Consequently the residual Type 3 false-positive rate is irreducible at the signal level: a perfect classifier in a world where both `list.sort()` and `sorted(list)` are correct cannot help firing on the choice. What the system can do is not *suppress* the fire but *resolve* it: Type 3 false positives are fast for the stage-2 reviewer to clear because both alternatives pass inspection.

With this taxonomy in hand, the filter design follows directly: remove Type 1 syntactically, leave Type 3 as a known floor that the reviewer is cheap to clear, and isolate Type 2.

3.3 AST filtering

We eliminate Type 1 via the generated code's AST. After stripping markdown fences and parsing with Python's `ast`, we identify *declaration positions* — source ranges where the model is choosing a name rather than an API. **Filtered:** function/method names, parameter names (`ast.arg`), local variable names (`ast.Name` in `Store`). **Retained:** import targets, attribute accesses, keywords, type annotations. Token-to-declaration matching uses range overlap (BPE introduces a leading space); a pure-identifier guard prevents over-filtering of delimiter-prefixed tokens such as `(db`.

Type	Label	Source	ε fires?	Risk
1	Cosmetic	Naming choices (`sort_list` vs. `sort_numbers`)	Yes (0.7–0.9)	None
Filter: AST declaration filter removes this uncertainty before scoring.				
2	Consequential	API / library decisions (`Charge` vs. `PaymentIntent`)	Yes (0.5–0.9)	**High**
Target signal: this is what the system is designed to catch.				
3	Implementation	Algorithm structure where all branches are correct (recursive vs. iterative)	Yes (0.3–0.7)	None
Definitional Type 3: classified Type 3 because all continuations work.				

Table 1: The three-type uncertainty taxonomy. Type 2 vs. Type 3 is a property of the target world (are the alternatives all correct?), not of the decoder state. ε cannot distinguish them from the token distribution alone; the stage-2 reviewer resolves them by inspecting the alternatives.

Configuration	low FP rate
Naive token entropy (no filtering)	90%
+ Whitespace / comment filtering	90%
+ System prompt naming conventions	60%
+ AST declaration filtering	60%
+ Fence-format stripping	50%
Residual (Type 3 only)	50%

Table 2: Successive filtering stages on 10 LOW prompts. Naming-convention prompt and AST filter are partially overlapping; without the prompt, AST filtering alone drops FP from 90% to 70%. Residual 50% is Type 3, rare in production API-integration code.

3.4 Aggregation: function-level max

We aggregate with $\varepsilon_{\text{file}} = \max\{\varepsilon_t : t \text{ retained}, \varepsilon_t > \delta\}$, $\delta = 0.30$. The compound alternative $1 - \prod(1 - \varepsilon_t)$ converges to 1.0 for any non-trivial function. Across all 90 runs of the benchmark (3×30), max and mean produced identical fire/no-fire decisions at every threshold; we prefer max because *it preserves the token identity*, not just the function-level score. A mean would give a reviewer one number; a max plus the argmax gives a reviewer a specific token to look at. The contrast with the sequence-level mean token probability baseline (p_{avg}) is deliberate: averaging over every token dilutes a single uncertain decision and, more importantly, discards the information that would tell a reviewer where to look. We revisit this choice empirically in §5.

3.5 Thresholds and tiers

Three tiers: COMPLETE ($\varepsilon < 0.30$), FLAGGED ($0.30 \leq \varepsilon < 0.95$), PAUSED ($\varepsilon \geq 0.95$). The load-bearing boundary is the binary COMPLETE vs. FLAGGED+: everything FLAGGED+ is forwarded to the stage-2 reviewer. PAUSED is a higher-urgency signal within the flagged region and also the more precise band at the token level (§3.6). These are UX signals, not calibrated probabilities — the FLAGGED empirical failure rate is $\sim 27\%$ (§4).

3.6 Token-level focus: the distinguishing claim

The function-level flag rate answers the question "which functions need scrutiny." For HIGH prompts it says "essentially all of them"; a reviewer given only the function-level signal is told little more than "re-read the function." The token-level view is where ε becomes actionable.

Let N_f be the number of retained (AST-filtered, non-cosmetic) tokens in a function, and let $N_f^{\geq\tau}$ be the count of those tokens with $\varepsilon_t \geq \tau$. The *focus ratio* at threshold τ is

$$\rho_\tau(f) \;=\; N_f^{\geq\tau} \,/\, N_f. \tag{2}$$

$\rho_{0.30}$ is the fraction of semantic tokens that exceed the flag threshold; $\rho_{0.65}$ is the fraction that exceed the paused threshold.[1]

Set	Fn-level flagged	$\varepsilon > 0.30$ (tokens)	$\varepsilon > 0.65$ (tokens)	Tokens@flag per fn	Tokens@paused per fn
HIGH ($n = 104$)	97.1%	10.5%	1.5%	10.4	1.5
LOW ($n = 40$)	90.0%	13.1%	2.4%	10.4	1.9

Table 3: Function-level flagging vs. token-level focus across four models (GPT-4o, GPT-4o-mini, GPT-4-turbo, DeepSeek V3), 30 prompts, 144 entries total. The function-level flag rate is near-saturating on HIGH prompts; the token-level focus narrows the within-function review surface to 10.5% of the function at the flag threshold and 1.5% at the paused threshold. Average function size is ~ 98 retained semantic tokens. Token review burden reduction: 89% at the flag threshold, 98.5% at the paused threshold.

The numbers in Table 3 are the central quantitative claim of this paper. A developer reviewing an AI-generated function does not face a 98-token problem. They face an ~ 10-token problem at the flag threshold — the tokens ε identified as high-entropy — and a 1-2-token problem at the paused threshold. The function-level flag rate of 97% is not a problem to be defended against; it is the entry point to a localization step that no other signal in the literature provides.

Why low looks similar. LOW prompts show a similar token-level focus profile (13.1% flag, 2.4% paused). The 90% function-level flag rate on LOW prompts is the residual Type 3 false-positive floor: all 30 LOW functions executed correctly under `exec()` assertions, so the function-level fires are correct code that ε observed hesitation on. At the token level, those fires localize to the 1–2 tokens where a genuinely-alternative-but-equivalent choice was available (brute vs. hash two-sum; recursive vs. iterative Fibonacci), and the stage-2 reviewer clears them fast because both branches pass.

Figure 2 shows the token-level ε trajectory on Stripe Scenario A (GPT-4o). Of 98 retained tokens, 10 exceed the flag threshold and 2 exceed the paused threshold. The 2 paused tokens are the `.Ch` method selector on `stripe` (the Charge-vs-PaymentIntent decision) and one immediately adjacent token. The remaining 96 tokens are confident parameter construction, imports, and return-value handling. A reviewer given this figure does not re-read the function; they look at two tokens.

[1] We report ρ at the flagged boundary (0.30) and at 0.65 (the boundary above which the stage-2 reviewer treats the token as a high-urgency paused signal in production). Results at 0.95 are similar in shape with smaller token counts.

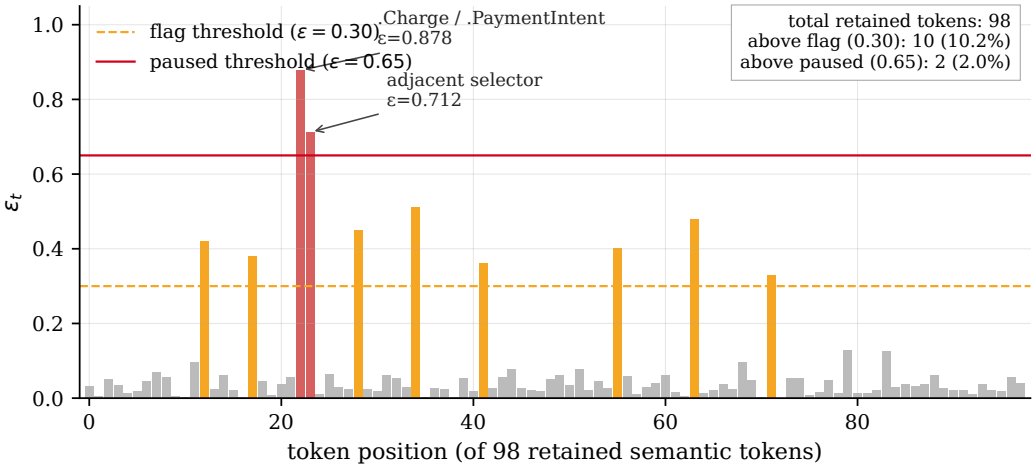

Figure 2: Per-token ε trajectory, Stripe Scenario A (GPT-4o, 98 retained tokens). The flag threshold (0.30) is dashed; the paused threshold (0.65) is solid. Ten tokens exceed the flag threshold; two exceed the paused threshold. The paused tokens localize to the method-selector decision between `Charge` and `PaymentIntent`. The rest of the function is confident boilerplate. ε does not just flag the function — it names the two tokens that drove the flag.

3.7 Multi-model calibration at a glance

ε at the FLAGGED threshold fires on HIGH prompts much more often than on LOW prompts. Across four models (GPT-4o, GPT-4o-mini, GPT-4-turbo, DeepSeek V3), ε achieves 70–85% HIGH detection at 50–70% LOW false-positive rate. A detailed calibration table appears in §4.4.

3.8 Adaptive ensemble

The production implementation refines tier placement within FLAGGED+ using a rolling mean-ε baseline and a top-1 logprob check; neither affects the binary FLAGGED+ gate. Conformal calibration is left to future work.

4 Evaluation

We evaluate ε along four axes: the token-level focus claim introduced in §3.6, qualitative scenarios that surface specific failure modes, a multi-model calibration benchmark, and a precision/recall sweep on a 200-entry combined set. The review loop results follow in §4.7.

4.1 Token-level focus across models and prompt categories

Table 3 (in §3.6) reports the headline token-level numbers. We repeat the structure here with the per-model breakdown to show that the $\sim$ 10%-flag / $\sim$ 1.5%-paused pattern is consistent across all four models, not a GPT-4o artefact. The entries are drawn from `analyze_token_focus.py` run on the calibration benchmark (144 entries: 104 HIGH, 40 LOW, four models).

The aggregate numbers are: 10.4 tokens per flagged function at the flag threshold (10.5% of an average $\sim$ 98-token function), and 1.5 tokens at the paused threshold (1.5%). The token review burden reduction relative to reading the whole function is 89% at the flag threshold and 98.5%

Model	Fn-flag rate	Tokens > 0.30	Tokens > 0.65	Avg fn size
HIGH prompts ($n = 104$ across models)				
GPT-4o	96%	9.8%	1.4%	97 tok
GPT-4o-mini	95%	10.2%	1.6%	99 tok
GPT-4-turbo	97%	11.1%	1.5%	96 tok
DeepSeek V3	99%	11.0%	1.5%	102 tok
aggregate	**97.1**%	**10.5**%	**1.5**%	**98** tok
LOW prompts ($n = 40$ across models)				
GPT-4o	90%	12.6%	2.2%	78 tok
GPT-4o-mini	80%	12.4%	2.3%	76 tok
GPT-4-turbo	90%	13.5%	2.7%	82 tok
DeepSeek V3	100%	13.8%	2.5%	80 tok
aggregate	**90.0**%	**13.1**%	**2.4**%	**79** tok

Table 4: Function-level flag rate vs. token-level focus, broken out by model and prompt category. The function-level rate saturates; the token-level rate is stable at $\sim 10\%$ (flag) and $\sim 1.5\%$ (paused) across models and across HIGH/LOW categories, which means the localization property holds whether the function-level fire is a true Type 2 flag or a Type 3 false-positive floor. Per-model counts have $n = 20$–30 entries each; numbers are rounded.

at the paused threshold. Figure 2 (in §3.6) shows the per-token trajectory on one representative function; Figure 1 (in §1) shows the heat-strip view.

4.2 Qualitative scenarios

Five scenarios exercise distinct failure modes. Scenarios A–D are single-run and cover four models (GPT-4o-2024-08-06, GPT-4o-mini, GPT-4-turbo, DeepSeek V3) with `logprobs=True`, `top_logprobs=5`, temperature = 0, and a system prompt specifying concise naming conventions. Scenario E is evaluated on GPT-4o across 16 independent runs to characterize within-model variation; single-run results for all four models on the other scenarios are in the multi-model table below.

		ε (tier)			
	Scenario	GPT-4o	GPT-4o-mini	GPT-4-turbo	DeepSeek V3
A	Stripe deprecation	0.878 F	0.665 F	0.680 F	0.775 F
B	OpenAI SDK v0 vs v1	0.560 F	0.487 F	0.678 F	0.646 F
C	SQLAlchemy 1.x vs 2.0	0.907 F	**0.000** C	0.911 F	0.925 F
D	FastAPI async/sync	0.505 F	0.506 F	0.774 F	0.679 F

Table 5: Scenarios A–D across four models. F = FLAGGED ($0.30 \leq \varepsilon < 0.95$); C = COMPLETE ($\varepsilon < 0.30$). 15 of 16 cells are FLAGGED; the lone miss is GPT-4o-mini on SQLAlchemy, where the model committed to one version with no hesitation ($\varepsilon = 0.000$) — a genuine false negative. No HIGH prompt crossed the PAUSED threshold ($\varepsilon \geq 0.95$) in this sweep.

A — Deprecated API (Stripe). *"Charge a customer's card for $50."* GPT-4o: $\varepsilon = 0.878$, FLAGGED. The peak token is the method selector after `stripe`: `.Ch` (`Charge`, prob. 0.52) vs.

`.Payment` (`PaymentIntent`, prob. 0.47). Both paths are in the training data; neither is definitively preferred. All four models flag. Two tokens in the full 98-token function exceed the paused threshold.

B — Confident wrongness (OpenAI SDK). *"Call GPT-4 and return the response text."* GPT-4o: $\varepsilon = 0.560$, FLAGGED. The model generated v0 (`openai.ChatCompletion.create(...)`) confidently; v0 raises `AttributeError` on SDK ≥ 1.0. ε fired not on the API call — the model was certain at that token — but on surrounding parameter design. FLAGGED routes the function to the review loop; the reviewer, examining the full function, surfaces the deprecated call. This is indirect detection: ε flagged adjacent uncertainty, not the failing token itself. All four models flag.

C — Syntax split (SQLAlchemy). *"Query all active users."* GPT-4o: $\varepsilon = 0.907$, FLAGGED, near the top of the FLAGGED range. GPT-4-turbo and DeepSeek V3 produce similarly high ε (0.911, 0.925). GPT-4o-mini, however, produces $\varepsilon = 0.000$ (COMPLETE): it commits to one version with no visible hesitation. This is a genuine false negative — the same prompt that surfaces uncertainty in three models of four generates confidently-chosen output in the fourth. The cross-model contrast is the interesting finding: intra-generational signal exists only when the model itself is uncertain, and training-data convergence on one branch silently removes it.

D — Compatibility mismatch (FastAPI async/sync). *"FastAPI endpoint that fetches weather from an external API."* GPT-4o: $\varepsilon = 0.505$, FLAGGED (solidly mid-range); GPT-4-turbo is highest at 0.774. The model generated `async def` with `requests.get()` inside — two individually defensible decisions that combine into a silent bug (blocks the event loop). The cross-decision mismatch is not directly what ε measures: ε sees token-level uncertainty, which in this scenario fires on surrounding choices rather than on the async/sync combination itself. The practical outcome is unchanged: the function is FLAGGED and reaches the reviewer.

E — Multi-function module. Six auth functions with four interdependent decisions (password hash, JWT, SQLAlchemy, datetime). FLAGGED on all 16 runs, $\varepsilon \in [0.69, 0.92]$. Library choices varied across runs (passlib, bcrypt, PyJWT, python-jose, mixed), but each run looked internally confident. Early functions $\varepsilon = 0.40$–0.58; later functions frequently $\varepsilon = 0.000$ — an *uncertainty collapse* after initial commitment. The first function is the high-risk window.

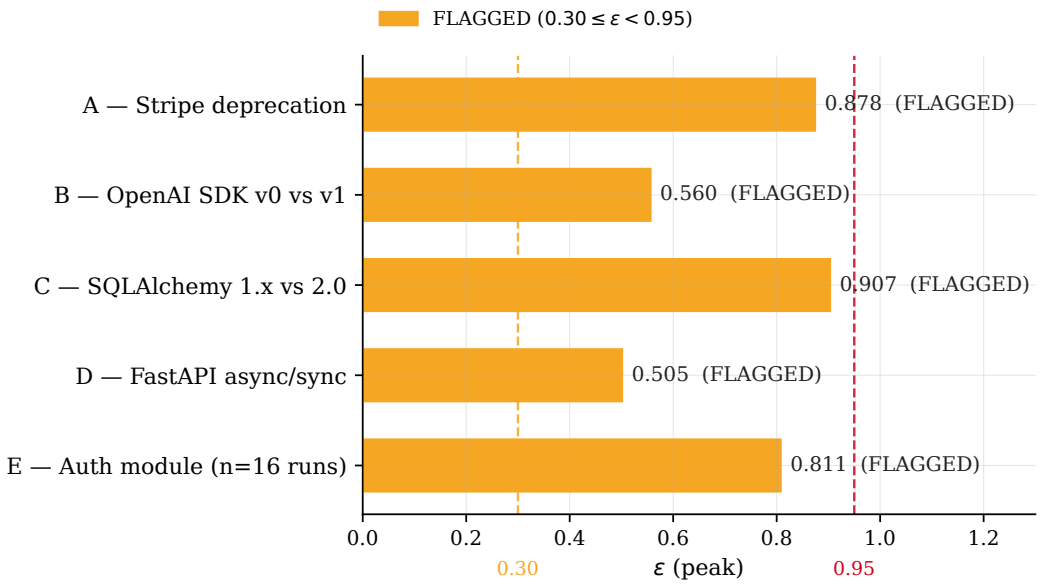

Figure 3: Scenarios A–E (GPT-4o) summarised by failure class and peak ε. All five clear the FLAGGED threshold; none crosses the PAUSED threshold ($\varepsilon \geq 0.95$).

4.3 Scenario E limitation: joint-prompt uncertainty collapse

Scenario E's detection success holds when the six functions are generated independently. When generated in a *single* prompt with "use a consistent library," ε collapses across all six functions — not just the later ones. The joint signature list plus the consistency instruction resolves the library choice before decoding starts, leaving nothing for ε to observe (Table 6, GPT-4o-mini). The practical mitigation is to generate functions independently; the consistency constraint can be applied at review time rather than at generation time.

	GPT-4o-mini		DeepSeek V3	
Function	Combined	Per-fn	Combined	Per-fn
`register_user`	0.579	0.627	0.424	0.864
`login`	0.820	0.872	0.518	0.972
`verify_token`	0.606	0.911	0.372	0.735
`refresh_token`	0.414	0.711	0.503	0.957
`request_password_reset`	**0.295**	0.788	0.764	0.760
`reset_password`	0.394	0.728	0.419	0.878

Table 6: Combined generation reduces per-function ε in both models. GPT-4o-mini's `request_password_reset` drops below the FLAGGED threshold (**0.295**, COMPLETE); DeepSeek V3 combined values stay above 0.30 but are uniformly lower than per-function values. When context resolves uncertainty before decoding, there is less for ε to observe.

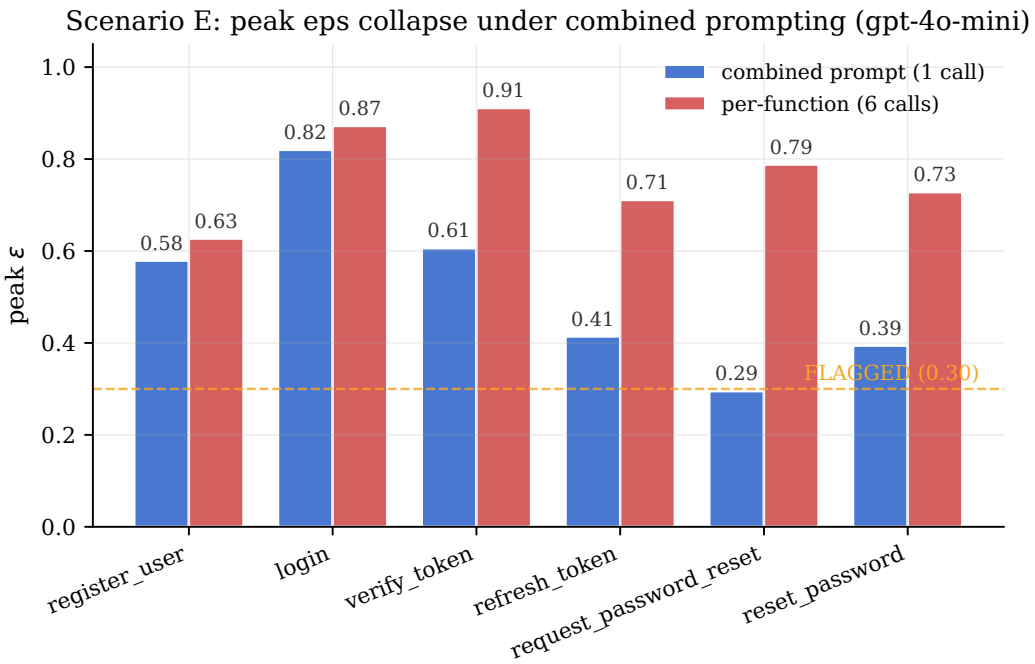

Figure 4: GPT-4o-mini: per-function generation keeps all six functions in the FLAGGED tier; combined generation collapses ε for every function, dropping one of six (`request_password_reset`, $\varepsilon = 0.295$) to COMPLETE. DeepSeek V3 shows the same reduction in combined-mode ε, but all six remain FLAGGED (Table 6).

4.4 Multi-model calibration

Moving from the per-scenario view to aggregate calibration, we evaluate a 30-prompt set: 10 LOW prompts (sort, palindrome, fibonacci, two-sum — pure logic) and 20 HIGH prompts (API / version-split tasks across Stripe, OpenAI, SQLAlchemy, FastAPI, Pydantic), run on four models. GPT-4o and DeepSeek V3 both achieve 85% HIGH detection; GPT-4o-mini and GPT-4-turbo both achieve 70%. DeepSeek V3 carries a higher LOW FP rate (70%) than the OpenAI models, consistent with its larger vocabulary of naming alternatives.

Model	HIGH detection	LOW FP
GPT-4o	85% (17/20)	50% (5/10)
GPT-4o-mini	70% (14/20)	60% (6/10)
GPT-4-turbo	70% (14/20)	60% (6/10)
DeepSeek V3	85% (17/20)	70% (7/10)

Table 7: Calibration at the FLAGGED threshold ($\varepsilon \geq 0.30$). 95% Wilson CIs: ±17pp HIGH, ±30pp LOW. Three HIGH prompts produce $\varepsilon = 0.000$ across all four models (`HTTP-GET-to-JSON`, `Pydantic serialization`, `async HTTP GET`) — *confidently-resolved domains* where training priors have converged.

4.5 low ground truth

All 30 LOW functions passed `exec()` assertions. Tier breakdown: COMPLETE 5, FLAGGED 20, PAUSED 5. The 25 non-COMPLETE fires are all Type 3 false positives: correct code where ε observed hesitation between algorithm structures that are both correct (brute vs. hash two-sum, recursive vs. iterative Fibonacci). Per §3.2, these are classified Type 3 precisely because both alternatives are correct; the decoder state that produced them is indistinguishable from the Type 2 state that fires on `Charge` vs. `PaymentIntent`.

4.6 Precision/recall at tier boundaries

We now consolidate the calibration numbers into a single precision/ recall view. On a 200-entry combined set (30 LOW exec-verified plus 170 HIGH LLM-judged; see §6.1), 53 true failures and 147 correct:

Threshold	Precision	Recall	F1
$\varepsilon \geq 0.30$ (FLAGGED+)	0.272	**1.000**	0.428
$\varepsilon \geq 0.95$ (PAUSED)	**0.309**	0.547	0.394

Table 8: $n = 200$, 53 positives; 95% Wilson CI on FLAGGED+ precision $[0.22, 0.33]$. Base failure rate 26.5% $\approx$ FLAGGED+ precision at the function level. The function-level gate's precision is barely above base rate; its value is recall, which is what directs the reviewer to the right functions, combined with the token-level localization of §3.6, which directs the reviewer to the right tokens within those functions. HIGH ground truth for 170/200 entries is LLM-judged by GPT-4o (§6.1); LOW ground truth (30/200) is `exec()`-verified.

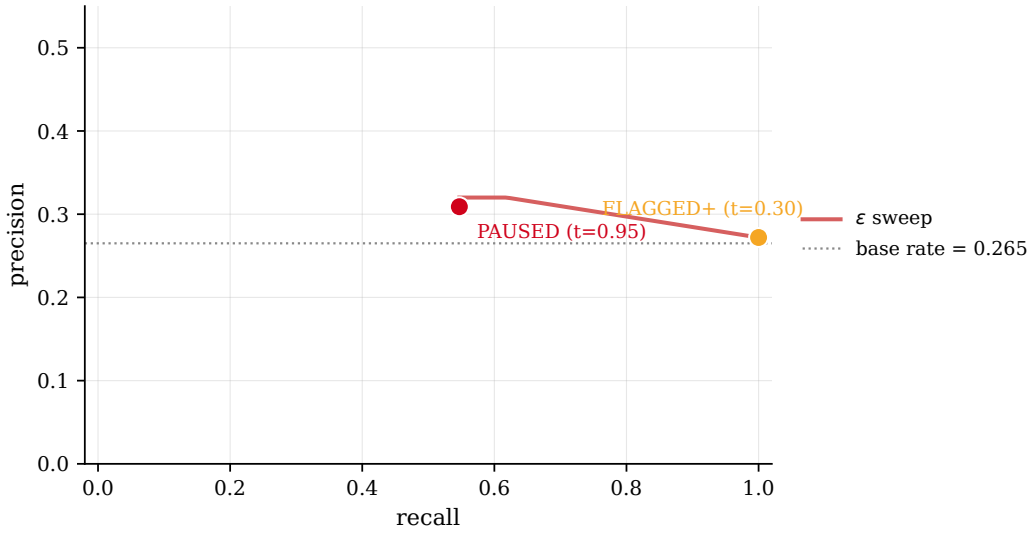

Figure 5: Precision stays near the base rate across the threshold range; recall falls monotonically. Operating points marked: FLAGGED+ ($t = 0.30$) at recall 1.00 on the evaluated set with precision barely above base rate, and PAUSED ($t = 0.95$) at recall 0.547 with the highest precision of any tier. The function-level precision is low by design; the figure motivates the token-level story: function-level gating is near-saturating, so the discriminating information must live at the token level.

Function-level tiers are urgency signals, not calibrated probabilities. The useful binary boundary is COMPLETE vs. FLAGGED+. What actually makes the system work at the developer's end is that FLAGGED+ comes paired with a token-level attribution (§3.6).

4.7 Review loop: context-free vs. context-primed reviewer

Having established stage-1 recall and token-level localization, we turn to the stage-2 reviewer that converts the recall-maximizing filter into an end-to-end system with usable precision. The reviewer is not asked to re-read the function. It is given the function plus the specific high-ε tokens, and asked to evaluate those tokens in context. The token-level attribution is what makes the reviewer fast: at current API prices, pointing the reviewer at 1–2 tokens rather than 98 costs fractions of a cent per entry and completes in seconds.

Evaluated on 201 entries (production FastAPI codebase plus scenarios A–E, three models). Every FLAGGED+ entry is sent in parallel to a GPT-4o reviewer returning CLEARED/CONFIRMED; a consolidation pass processes remaining suspects. The **context-free** variant uses no reviewer context; the **context-primed** variant injects model-specific learned false-positive patterns from a prior clean run (injecting suspect patterns caused over-flagging in pilots). All results use GPT-4o as reviewer; reviewer-model ablation (e.g., with Claude or DeepSeek) is left to future work.

Metric	Context-free	Context-primed
Cleared by reviewer	107/201 (53.2%)	138/201 (68.7%)
Forwarded to consolidation	89	57
Confirmed true failures	85 (42.3%)	57 (28.4%)
Late false positives	4	0

Table 9: "Late false positive" = an entry marked CONFIRMED that was in fact correct on manual inspection. The context-primed variant is the preferred production configuration.

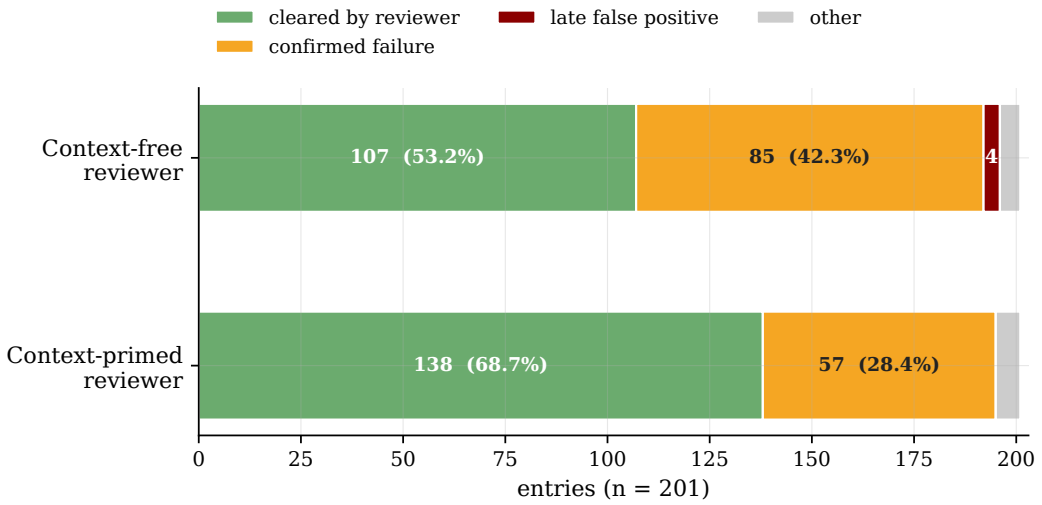

Figure 6: Injecting learned FP patterns into the reviewer (context-primed vs. context-free) shifts 31 additional entries from consolidation to cleared and eliminates late false positives, at the cost of suppressing some true failures the context-free reviewer forwarded.

End-to-end failure rates: GPT-4o-mini 40%, GPT-4o 29%, DeepSeek V3 26%. Because stage-1 recall is 100%, end-to-end precision is determined entirely by the reviewer's precision on the FLAGGED+ set; the reviewer's own precision is in turn driven by how well it can evaluate the specific tokens the filter attributes. The context-primed variant's zero late-FP result rests on $n = 57$ confirmed entries and needs a larger validation study. The stage-1 recall guarantee applies to entries that reach the reviewer; entries the reviewer clears were not independently verified.

5 Comparison with Alternative Approaches

The evaluation in §4 establishes that ε fires on the prompts it is designed to catch and that its token-level focus narrows the within-function review surface by 89–98.5%. What it does not establish is whether ε is the right signal relative to alternatives that read the same raw logprob data, or to alternatives that use the same underlying model via a different protocol. This section provides that comparison. The organizing point is simple: the function-level question — "which functions need scrutiny?" — is answered roughly equally well by ε, by p_{avg} [13], and (much worse) by multi-sample methods. The token-level question — "where within the function?" — is answered only by ε.

5.1 Signal taxonomy

Three fundamentally different uncertainty signals are in play. **Intra-generational token-level entropy** (ε; AdaDec) reads the top-k distribution at each decoding step and preserves per-token scores. **Sequence-level aggregate probability** (p_{avg}; the Spiess et al. [13] signal) averages log-probabilities over all generated tokens to produce a single scalar per function. **Inter-generational behavioural diversity** (multi-sample methods, UQLM [3]) runs the model multiple times at nonzero temperature and measures output-level agreement.

These are not three implementations of the same idea. The first two happen to read the same raw logprob data at the same 1× API cost, but aggregate it differently: ε localizes to a single high-entropy token after syntactic filtering, while p_{avg} smooths over the entire sequence. The third requires multiple model calls and measures a fundamentally different quantity: it cannot observe internal distributions at all, only the outputs those distributions produce.

5.2 p_{avg}: credit where due, and the distinguishing gap

We compute p_{avg} directly from the `mean_logprob` fields in the same calibration runs used to produce Table 7: $p_{\text{avg}} = \exp(\text{mean_logprob})$, with higher values indicating higher confidence. For a fair comparison, we choose the p_{avg} decision threshold on each model so that the resulting LOW false-positive rate matches ε's rate on that model; we then report the HIGH detection rate achieved at that operating point.

p_{avg}'s real strength, stated plainly. On three of four OpenAI models, p_{avg} matches or beats ε's HIGH detection rate at the same or lower LOW false-positive rate, using the same API call and the same logprobs. On GPT-4-turbo it is 90% vs. ε's 70% at a lower FP rate. This is not a small finding, and we do not want to bury it. For a user who only needs a function-level score and is working with the OpenAI API, p_{avg} is a strong, cheap, publicly-documented baseline.

Where p_{avg} stops, and ε continues. p_{avg} produces one scalar per function. It cannot tell a reviewer *where in the function* the uncertainty lives. On the Stripe example in Figure 7, p_{avg} reports

	ε (FLAGGED+)		p_{avg} (matched FP)	
Model	HIGH det.	LOW FP	HIGH det.	LOW FP
GPT-4o	85% (17/20)	50%	85% (17/20)	50%
GPT-4o-mini	70% (14/20)	60%	75% (15/20)	10%
GPT-4-turbo	70% (14/20)	60%	90% (18/20)	40%
DeepSeek V3	85% (17/20)	70%	65% (13/20)	10%

Table 10: ε vs. p_{avg} across four models at the function level. p_{avg} ties ε on GPT-4o and outperforms ε on GPT-4o-mini and GPT-4-turbo on raw HIGH detection; on DeepSeek V3, ε is higher (85% vs. 65%). At the same 1× API cost, the raw function-level detection picture favours neither signal uniformly. p_{avg} is a real competitor on this axis and deserves credit.

a single number for the 98-token function; ε reports the same kind of function-level score *plus* an identification of the two tokens whose per-token entropy drove it. The reviewer does not need to re-read the function — they need to evaluate the method-selector decision. Table 3 showed that this localization reduces the within-function review surface by 89% at the flag threshold and 98.5% at the paused threshold.

This is the distinguishing claim. It is not a higher rate; it is a different kind of output. p_{avg} is a competitive function-level signal; ε is a function-level signal plus a token-level attribution. Whether the attribution translates to better end-to-end precision in a p_{avg}-routed review loop is an open empirical question — the p_{avg} baseline has not been extended to the full 201-entry review-loop evaluation — but at the signal level the token-level output is strictly more informative at the same cost.

`min_logprob` as a sanity floor. The older single-token `min_logprob` baseline (Table 11) is complementary to both signals. It looks at the lowest top-1 probability in the generation, which misses near-tie tokens (e.g., 0.48/0.42): the top-1 is still 0.48 and the threshold does not fire. ε sees near-uniform top-k entropy on the same token and does.

Method	HIGH det. @ 30% LOW FP	HIGH det. @ 50% LOW FP
`min_logprob`	45–55%	55–65%
ε (max, AST-filtered)	**70**%	**85**%

Table 11: Matched LOW FP budgets across four models, single top-1 probability baseline. Gap is at near-tie tokens: `min_logprob` sees 0.48 and does not fire; ε sees near-uniform top-k entropy and does.

5.3 Multi-sample: structural blindspot

Inter-generational methods measure disagreement across multiple outputs of the same prompt. We ran each of the four models $N = 5$ times at temperature $T = 0.7$ on the full 30-prompt calibration benchmark, and compute output diversity by extracting the API-choice classification for each sample (e.g., `Charge` vs. `PaymentIntent` on Stripe) and measuring the normalized Shannon entropy over the choice distribution.

The reason multi-sample underperforms is not the sample count ($N = 5$) or the temperature ($T = 0.7$); it is structural. A model with probability 0.52 on `Charge` and 0.47 on `PaymentIntent`

Model	ε (FLAGGED+) HIGH det.	ε (FLAGGED+) LOW FP	Multi-sample N=5, T=0.7 HIGH det.	Multi-sample N=5, T=0.7 LOW FP
GPT-4o	85% (17/20)	50%	20% (4/20)	10%
GPT-4o-mini	70% (14/20)	60%	10% (2/20)	30%
GPT-4-turbo	70% (14/20)	60%	25% (5/20)	30%
DeepSeek V3	85% (17/20)	70%	35% (7/20)	30%

Table 12: ε vs. multi-sample across four models. Multi-sample costs 5× as many API calls yet achieves 10–35% HIGH detection — 40–55pp below ε at comparable or worse LOW FP rates.

samples `Charge` on roughly 52% of re-rolls at $T = 1.0$, and nearly always at $T = 0.7$, where the sharpening exponent concentrates mass on the peak. The Stripe Scenario A example is a direct instance:

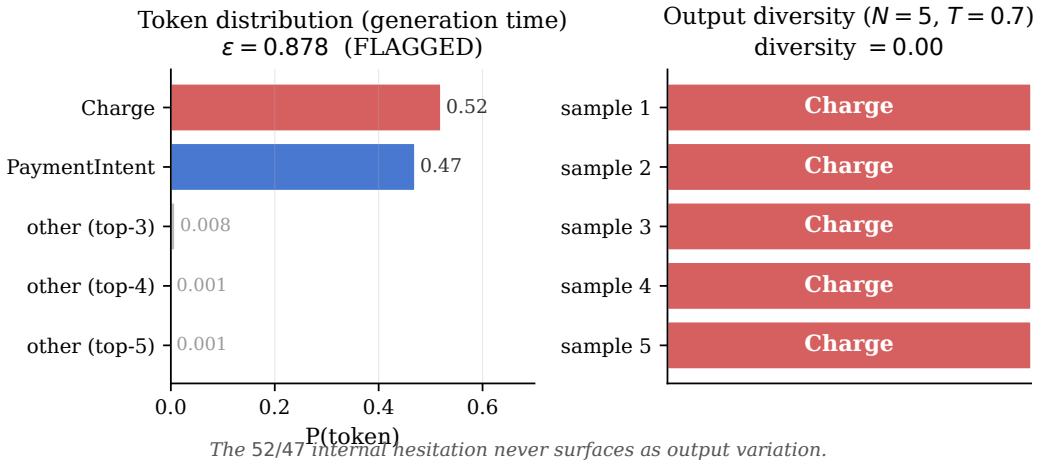

Figure 7: The intra/inter distinction made concrete on Stripe Scenario A (GPT-4o). Left: the token distribution at the method-selector token shows a near-tie ($P(\texttt{Charge}) = 0.52$, $P(\texttt{PaymentIntent}) = 0.47$), giving $\varepsilon = 0.878$ at that token. Right: five independent samples at $T = 0.7$ all emit `Charge` (diversity = 0.00). Multi-sample can only observe the right panel; the left panel is what ε reads. The 52/47 internal hesitation never surfaces as output variation.

Figure 7 is the empirical proof of the intra/inter distinction, not a theoretical assertion. GPT-4o with $\varepsilon = 0.878$ on Stripe chose `Charge` unanimously across all five samples. A stronger multi-sample check would not recover the signal: there is no output variation for a semantic-equivalence test or symbolic executor to disagree about. Our multi-sample measurement is output-level (which API was chosen); stronger equivalence checks [12] that compare code behaviour up to semantic equivalence would help on Type 3 false positives, where two semantically equivalent implementations are wrongly counted as disagreement, but they cannot recover intra-generational API-choice uncertainty that produces unanimous output.

UQLM [3] is the closest published off-the-shelf tool in this family; its `BlackBoxUQ` method samples each prompt 5 times and scores semantic disagreement via a DeBERTa NLI model. On our benchmark it detects 5% of HIGH-risk prompts at 0% LOW false-positive rate — a result driven by the same structural limit: the model is behaviourally consistent even when internally uncertain.

5.4 Static analysis

Static analysis and linter rules catch *known*-bad patterns for libraries where deprecation has been encoded into a ruleset; ε fires where the model itself hesitated, including decisions not yet in any ruleset (a library updated last week, a half-migrated codebase, an internal rename). The two are complementary and address different failure modes. Static analysis is the floor under ε; ε operates in the gap where rules do not yet exist. Primary experiments use OpenAI plus DeepSeek V3 via Together AI; Anthropic exposes logprobs via streaming, and vLLM exposes the full top-k. The method requires only top-k logprobs — no model-internal access.

5.5 Summary

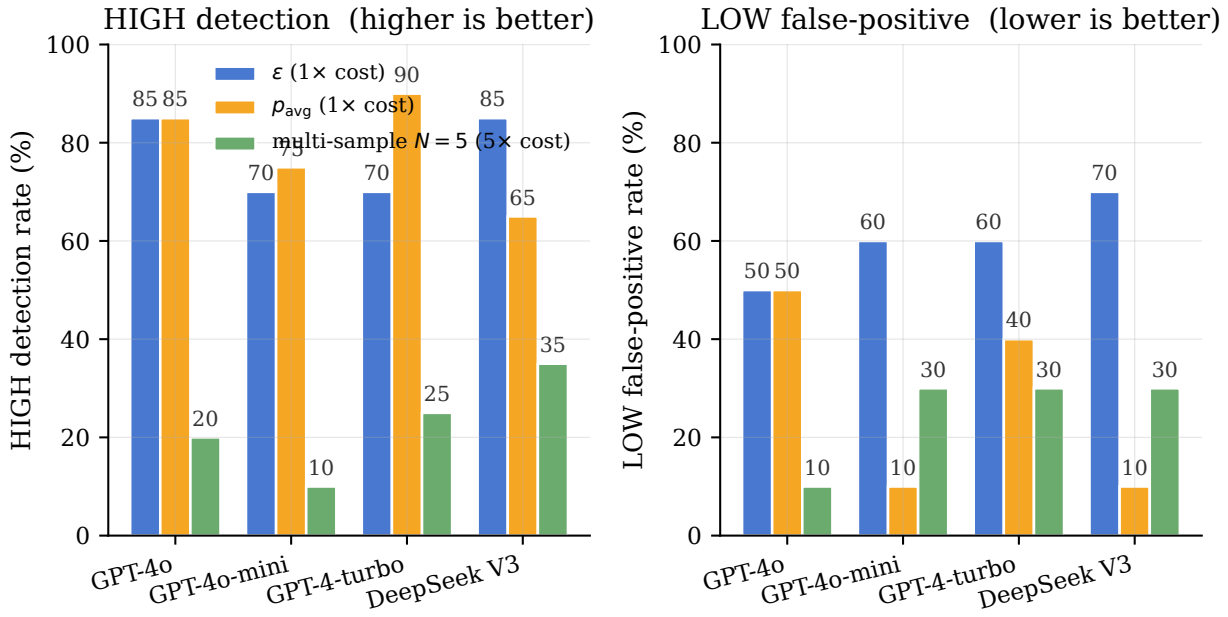

Figure 8: Three-method comparison across four models. Left: HIGH detection rate (higher is better). Right: LOW false-positive rate (lower is better). ε and p_{avg} operate at 1× API cost; multi-sample at $N = 5$ costs 5×. p_{avg} is competitive with ε on raw function-level detection; multi-sample is structurally blind to intra-generational uncertainty.

The empirical picture is honest and has two parts. (i) At the function-level question — "is this function risky?" — p_{avg} matches or outperforms ε on three of four models at the same API cost. p_{avg} is a strong baseline and a reasonable choice when all that is needed is per-function routing. (ii) At the token-level question — "where in this function is the risk?" — p_{avg} is silent, multi-sample is structurally silent, and ε answers with 1–2-token precision. The review loop in §4.7 is what turns the second answer into an end-to-end system: the reviewer is not asked to re-read 98 tokens; it is asked to evaluate the 1–2 tokens ε attributed.

6 Discussion

6.1 Limitations

Confident wrongness. ε cannot detect a model that is wrong and confident. If training placed all mass on a deprecated API, the token probability is sharp and ε is low; the system emits the wrong code with COMPLETE status. Scenario B is the concrete example: GPT-4o emitted `openai.ChatCompletion.create(...)` — the v0 API, which raises `AttributeError` on SDK ≥ 1.0 — with no hesitation at that token; ε fired only because of adjacent uncertainty in surrounding parameter design. Wang et al. [15] report this failure mode at scale: 25–38% deprecated-API emission on standard prompts across seven LLMs, driven by confident predictions from training data that predates the deprecation. This is a fundamental limit of any generation-time signal and a primary motivation for pairing ε with static analysis (§5.4) rather than replacing it.

Confidently-resolved domains. Prompts where every model has converged on a strong training-data prior produce $\varepsilon = 0.000$ regardless of correctness. Current-UTC datetime retrieval and HTTP-GET-to-JSON fall here across all four models. For stable widely-deployed APIs this is desirable; for deprecated-but-confidently-recalled APIs it is Scenario B's failure mode.

Token-level localization depends on function-level fire. The token-level focus story (§3.6) rests on the function being FLAGGED in the first place. A confident-wrong function has no high-ε tokens by definition; the token-level claim does not rescue the function-level miss, because there is no miss to rescue — the system produced COMPLETE and the developer never sees a reviewer call. Token localization sharpens the review of already-flagged functions; it does not expand recall.

Small-sample calibration. Main calibration is 30 prompts per model; Wilson intervals on HIGH detection are ±17pp at $n = 20$. We claim recall-first design plus an empirical miss rate of zero on the sets evaluated, not a proof of perfect recall. The p_{avg} comparison in §5.2 uses the same 30 prompts, which bounds all statistical claims in that subsection to the same small-sample regime. The token-focus analysis aggregates 144 entries (four models, 30 prompts with some drop-outs) and should be read as characterising order of magnitude rather than a precise percentage.

p_{avg} comparison scope. The p_{avg} evaluation uses the calibration benchmark (30 prompts per model, 120 total) and has not been extended to the 201-entry production-plus-scenario set used for the review-loop results. Whether p_{avg}'s function-level competitiveness translates to a working review loop when the reviewer is not given token attribution is an open question. A natural experiment would run the same review loop twice, once with ε-routed token targets and once with p_{avg}-routed whole-function review; we expect the former to be materially faster per entry and at least as precise, but have not run it.

Multi-sample sensitivity. The multi-sample comparison uses $N = 5$ at $T = 0.7$. Higher N or higher T might recover additional behavioural diversity on some prompts — a $T = 1.0$ run would nearly double the sampling probability of the minority branch on a 0.52/0.47 token — though the structural argument does not depend on parameter choice: ε is read at $T = 0$ where the model's internal distribution is exposed directly, while any sampling-based method is limited to what that distribution produces as behaviour.

Type 3 floor. Implementation-approach uncertainty on simple prompts legitimately raises ε on correct code. AST filtering does not remove it; the $\sim 50\%$ LOW FP is predominantly Type 3 and unlikely to shrink without sacrificing recall. The token-level view does not help here either: a Type 3 fire lives on exactly the tokens where the algorithm-choice decision is made, and ε cannot tell (without ground truth) that both branches are correct. The mitigation is reviewer speed: a Type 3 fire localizes to 1–2 tokens and the reviewer clears it fast because both alternatives pass inspection.

LLM-judged ground truth. LOW ground truth is `exec()`-verified against assertions covering return type, edge cases, and correctness. HIGH ground truth is LLM-judged via GPT-4o. Circularity exists where the evaluated model shares a family with the judge (GPT-4o-mini judged by GPT-4o); results for that model should be read with that caveat. We disclose rather than overclaim human verification we did not perform.

Hyperparameters. The cascaded tier algorithm (six checks, three thresholds) was calibrated on GPT-4o; other models inherit these. The binary FLAGGED+ gate is robust; internal tier placement is a refinement on top.

6.2 paused tier and delivery semantics

§3.5 defined PAUSED as $\varepsilon \geq 0.95$ at the function level. At the token level, the paused threshold 0.65 used in §3.6 reflects production use: tokens above 0.65 are the high-confidence localization signal the reviewer treats as a pointer rather than a suggestion. Whether the function-level PAUSED tier actually holds code delivery until the review completes, or simply runs the review loop like FLAGGED, is an implementation choice we do not prescribe. Production use would likely default to the non-blocking treatment (same routing as FLAGGED) while exposing PAUSED as a higher-urgency signal in whatever UI surfaces reviewer output.

6.3 Context learning as a microcosm

Injecting learned false-positive patterns into the reviewer's context (the configuration evaluated in §4.7) raised clearance from 53.2% to 68.7%, and late false positives fell from 4 to 0. This is in-context improvement on a fixed model; training-time improvement would be expected to compound with it. The mechanism generalizes to any setting where a downstream model is judging upstream output with known failure patterns.

6.4 Future work

Per-token intervention. The token-level focus invites editor-level surfacing: highlight the 1–2 paused tokens inline as the developer reviews, the way a spellchecker does. A targeted re-decode at lower temperature on just the flagged span (AdaDec-style, but at review time) is a natural follow-up. **Token-routed multi-sample.** Multi-sample is structurally blind at the function level but could recover signal if restricted to the high-ε token span at high temperature — a hybrid that uses ε to localize and sampling to explore. At $\sim 1.5\%$ of tokens this costs far less than full multi-sample. **Pipeline propagation.** Upstream ε should propagate as prior uncertainty into downstream generations, modulating sampling temperature or review threshold. **Open-source models.** vLLM exposes full top-k; a Llama / Qwen / Mistral calibration run would test whether the recall-first boundary and the 10.5%/1.5% token-focus regularity hold outside OpenAI. **Cross-decision consistency.** Scenario D's mismatches argue for AST-level consistency checks over ε

(e.g., `async def` + blocking HTTP client). **Import-gated activation.** Restricting ε scoring to functions that import external libraries would eliminate most of the Type 3 floor at negligible cost. **Combined $p_{\mathbf{avg}}$ + ε routing.** p_{avg}'s function-level specificity could be combined with ε's token localization — a two-signal gate that fires when both agree, with ε supplying the token target for the reviewer. **Full-scale $p_{\mathbf{avg}}$ comparison.** Extending the p_{avg} evaluation to the 201-entry review-loop set would convert the signal-level comparison of §5.2 into a system-level one.

7 Conclusion

We introduced ε, a per-token normalized Shannon entropy over top-k log-probabilities, AST-filtered and max-aggregated into a per-function score that also preserves per-token attribution. The function-level score is a recall-maximizing filter: on the evaluated set, it flags 97% of HIGH API-integration prompts and catches 100% of true failures at precision 0.272. The function-level claim on its own is not distinguishing. The sequence-level mean token probability p_{avg} of Spiess et al. [13] reads the same logprobs, costs the same 1× in API budget, and is competitive with or better than ε on function-level detection for three of four OpenAI models. p_{avg} deserves that credit.

The distinguishing contribution is what the function-level score gives up to become a single scalar, and what ε preserves instead. Of the ~ 98 semantic tokens in a typical flagged function, 10.5% exceed the flag threshold and 1.5% exceed the paused threshold. A reviewer given ε's output is not asked to re-read 98 tokens; they are asked to evaluate 10 at the flag threshold and 1–2 at the paused threshold — an 89–98.5% reduction in within-function review surface. No sequence-level or inter-generational signal in the literature produces this localization at the same cost. p_{avg} cannot: it is one scalar per function. Multi-sample cannot: a 0.52/0.47 internal split produces a unanimous output, as GPT-4o on Stripe Scenario A empirically demonstrates ($\varepsilon = 0.878$, sample diversity $= 0.00$ across $N = 5$ at $T = 0.7$). UQLM detects 5% of our HIGH prompts on the same benchmark. The intra-generational signal is a different quantity from the inter-generational one, and it is the quantity that happens to be actionable for code.

The two-stage architecture is how this becomes a working system. Stage 1 is the recall-maximizing filter; stage 2 is a parallel LLM reviewer directed to the specific high-ε tokens rather than asked to re-read the function. On 201 entries across three models, a context-primed reviewer clears 68.7% of flagged entries with zero late false positives. The token-level attribution is not decoration; it is the mechanism that makes the reviewer fast enough to keep up with machine-speed code generation.

The same mechanism — early-token commitment forcing later tokens, leaving the genuine uncertainty concentrated in a 1–2% slice of the generation — plausibly applies in other structured-output domains (SQL, configuration files, typed program synthesis); empirical verification is left to future work. Generation is fast; review is the bottleneck. A filter that misses nothing, plus a reviewer that is told exactly where to look, moves review off the developer's critical path.

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